

3a) Data Types:

Write a Java program to store and display the following details of a waste collection vehicle:

Vehicle number - integer

Waste collected in kilograms - decimal value

Number of collection points - integer

Vehicle status - character

Use appropriate Java data types for each value and display all the details.

```
import java.util.Scanner;

public class Vehicle {
    public static void main(String[] args) {
        Scanner nope = new Scanner(System.in);

        System.out.print("Please enter Vehicle Number: ");
        int veh = nope.nextInt();

        System.out.print("Please enter weight of waste collected: ");
        double wei = nope.nextDouble();

        System.out.print("Please enter no. of collection points: ");
        int col = nope.nextInt();

        System.out.print("Please enter vehicle status: ");
        char sta = nope.next().charAt(0);

        System.out.printf("Vehicle number is: %s", veh);
        System.out.println(" ");
        System.out.printf("Weight of waste collected is: %s", wei);
        System.out.println(" ");
        System.out.printf("No. of collection points is: %s", col);
        System.out.println(" ");
        System.out.printf("Vehicle status is: %s", sta);
        System.out.println(" ");

    }
}
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>javac Vehicle.java

C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>java Vehicle
Please enter Vehicle Number: 1234
Please enter weight of waste collected: 182.42
Please enter no. of collection points: 510
Please enter vehicle status: A
Vehicle number is: 1234
Weight of waste collected is: 182.42
No. of collection points is: 510
Vehicle status is: A

C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>
```

3b) If-Else Condition:

Write a Java program to check the waste collection status based on the amount of waste collected. Read the waste collected in kilograms.

If the waste collected is 100 kg or more, display "Collection Target Achieved".

Otherwise, display "More Waste Collection Required".

Use an if-else statement.

```
import java.util.Scanner;
```

```
public class Target {  
    public static void main(String[] args) {  
        Scanner nope = new Scanner(System.in);  
  
        System.out.print("Please enter the waste collected in kilograms: ");  
        double waste = nope.nextDouble();  
  
        if (waste >= 100) {  
            System.out.println("Collection Target Acheived.");  
        } else {  
            System.out.println("More Waste Collection Required.");  
        }  
    }  
}
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>javac Target.java
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>java Target  
Please enter the waste collected in kilograms: 19.53  
More Waste Collection Required.
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>java Target  
Please enter the waste collected in kilograms: 101.032  
Collection Target Acheived.
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>|
```

3c) Methods:

Write a Java program to calculate the total waste collected from two collection points using a method.

Create the following method:

calculateTotalWaste(double point1Waste, double point2Waste)

The method should return the total waste collected. Read the waste collected at the two collection points from the user, call the method, and display the total waste collected.

```
import java.util.Scanner;

public class Calculate {
    public static void main(String[] args) {
        Scanner nope = new Scanner(System.in);

        System.out.print("Please enter waste collected at point 1: ");
        double one = nope.nextDouble();

        System.out.print("Please enter waste collected at point 2: ");
        double two = nope.nextDouble();

        System.out.println("The total waste collected is: " +
calculateTotalWaste(one, two));
    }

    static double calculateTotalWaste(double point1Waste, double point2Waste) {
        return (point1Waste + point2Waste);
    }
}
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>javac Calculate.java
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>java Calculate
```

```
Please enter waste collected at point 1: 192.42
```

```
Please enter waste collected at point 2: 368.38
```

```
The total waste collected is: 560.8
```

```
C:\Users\Rishitha\OneDrive\Desktop\College stuff!!!\ProjectJ>|
```